

Template

1 Single-choice questions

1.1 Which of the following would you apply unsupervised learning to?

- A. Compress a high-resolution image to a low-resolution image.
- B. Develop a model to predict the stock market.
- C. Given the data of body dimensions collected from 1,000 consumers, determine the size of the clothes to be produced.
- D. Given a large dataset of medical records of patients including the disease name, predict the disease of the new patient by the symptom.

1.2 Which one is a non-parametric method?

- A. Linear regression with nonlinear basis functions
- B. K-Nearest Neighbors classifier
- C. Linear SVM
- D. Neural network

1.3 Which statement about neural networks is correct?

- A. A neural network with only one nonlinear hidden layer does not have the universal approximation ability.
- B. The size of the output cannot be larger than the size of the input.
- C. A neural network without nonlinear activation functions is equivalent to a linear model.
- D. CNN cannot contain a fully connected layer.

1.4 Suppose we have 10 models from 10 different datasets. For a new test sample $(x, y) = (4, 9)$ sampled from the same distribution that generated the training sets, we suppose $t(x = 5) = 8.5$ (ground truth), and ϵ (noise) is instantiated as 0.5. Suppose the predictions for this new test sample given by the 10 trained models are 8, 7, 11, 6, 5, 14, 10, 9, 11, 6, respectively. Based on these 10 trials, which of the following results is incorrect?

- A. The empirical mean squared error (MSE) is 7.3.
- B. The bias² is 0.04.
- C. The variance is 7.1.

2.1 In a convolutional layer of CNN, suppose the size of the input is $w \times h \times c$ (c is the number of channels), the filter size is $a \times b \times c$, the stride is u , and the pad is v .

- (1) Compute the size of the feature map (the output of the filter). (3 points)
- (2) Suppose that we have k different filters. Compute the number of parameters in the layer. (3 points)
- (3) Suppose the input is a gray-scale image, which is shown by a matrix $\mathbf{I}$:

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

There are two filters $\mathbf{F}_1$ and $\mathbf{F}_2$:

$$\mathbf{F}_1 = \begin{bmatrix} -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad \mathbf{F}_2 = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}.$$

Suppose the bias terms for the two filters are both zero. Compute the outputs of the two filters and the outputs of max-pooling. (4 points)

2.2 Suppose that a random variable $\mathbf{x}$ follows the Gaussian mixture distribution with $p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$. Where $\boldsymbol{\Theta} = \{\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}\}$, $\boldsymbol{\pi} = \{\pi_1, \dots, \pi_K\}$, $\boldsymbol{\mu} = \{\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K\}$, $\boldsymbol{\Sigma} = \{\boldsymbol{\Sigma}_1, \dots, \boldsymbol{\Sigma}_K\}$ and latent variable $z \sim \text{Categorical}(\boldsymbol{\pi})$, where $\boldsymbol{\pi} \geq 0$, $\sum_k \pi_k = 1$.

- (1) Given a dataset $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$, please write the log likelihood expression $\ln p(\mathcal{D}; \boldsymbol{\Theta})$. (1 point)

- (2) **Likelihood Decomposition.** Show that

$$\ln p(\mathcal{D}; \boldsymbol{\Theta}) \geq \mathcal{L}(\mathbf{q}; \boldsymbol{\Theta}), \forall \mathbf{q}, \boldsymbol{\Theta},$$

where

$$\mathcal{L}(\mathbf{q}; \boldsymbol{\Theta}) = \sum_{n=1}^N \mathbb{E}_{q_n(z^{(n)})} \left[\ln \left(\frac{p(\mathbf{x}^{(n)}, z^{(n)}; \boldsymbol{\Theta})}{q_n(z^{(n)})} \right) \right],$$

and $\mathbf{q}(\mathbf{z}) = \prod_{n=1}^N q_n(z^{(n)})$. And, write down the gap between $\ln p(\mathcal{D}; \boldsymbol{\Theta})$ and $\mathcal{L}(\mathbf{q}; \boldsymbol{\Theta})$. (6 points)

- (3) **E-Step Derivation.** Given $\boldsymbol{\Theta} = \{\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}\}$, update $\mathbf{q}(\mathbf{z})$. (4 points)
- (4) **M-Step Derivation** Given $\mathbf{q}(\mathbf{z})$, update $\boldsymbol{\Theta} = \{\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}\}$. (6 points)
- (5) Please prove that $\ln p(\mathcal{D}; \boldsymbol{\Theta}^{t+1}) \geq \ln p(\mathcal{D}; \boldsymbol{\Theta}^t)$ holds for any t , with t being the index of one (E-Step, M-Step) iteration. (3 points)

2 Calculations and Derivations